

Jaykrishnan G.

Optimization Specialist

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Research Interests

- Theoretical Computer Science.
- Operations Research.

Education

The Technion – Israel Institute of Technology

DOCTOR OF PHILOSOPHY

Advisor: Prof. Asaf Levin

Haifa, Israel

JULY 2021 – DECEMBER 2024

The Technion – Israel Institute of Technology

MASTER OF SCIENCE

Advisor: Prof. Asaf Levin

Haifa, Israel

MARCH 2019 – JULY 2021

National Institute of Technology

MASTER OF TECHNOLOGY

Tamil Nadu, India

JUNE 2013 – MAY 2015

University of Kerala

BACHELOR OF TECHNOLOGY

Kerala, India

MAY 2008 – JULY 2012

Professional Experience

Optym

OPERATIONS RESEARCH SCIENTIST

JANUARY 2025 – PRESENT

- Contributed to the development and rollout of Optym's state-of-the-art railway scheduling system.

Publications

1. G. Jaykrishnan and A. Levin. EPTAS for load balancing problem on parallel machines with a non-renewable resource. *Discrete Optimization*, 48:100775, 2023
2. G. Jaykrishnan and A. Levin. EPTAS for parallel identical machine scheduling with time restrictions. *Journal of Combinatorial Optimization*, 47(2):10, 2024
3. G. Jaykrishnan and A. Levin. EPTAS for the dual of splittable bin packing with cardinality constraint. *Theoretical Computer Science*, 979:114202, 2023
4. G. Jaykrishnan and A. Levin. Scheduling with cardinality dependent unavailability periods. *European Journal of Operational Research*, 316(2):443–458, 2024
5. G. Jaykrishnan and A. Levin. APTAS for bin packing with general cost structures. *arXiv preprint arXiv:2407.07677*, 2024

Technical skills

- Languages: Python, C++, C#, LaTeX
- Libraries: CVXPY, CVXOPT, PyTorch, PyG, DGL, PyTorch Lightning, Hydra
- MLOps: Git, DVC, GitHub, Docker, Google Cloud Platform, CI/CD